

effort and adherence to industry-specific safety standards.

Ecosystem and tools: Consider the availability of development tools, libraries, and drivers that match your project's requirements.

Familiarity: If your team is already experienced with Linux development, Real-Time Linux with PREEMPT_RT patch, kernel config options, IRQ handling and thread priorities may be a more seamless choice. SCHED_FIFO, SCHED_RR are often used for Real-Time threads. These factors will improve real-time capabilities in the Linux environment.

Project scope: The size and complexity of the project may influence which platform is more suitable.

In conclusion, both QNX and Real-Time Linux are good options for developing real-time and embedded systems; each comes with its own strengths and considerations. The choice depends on the specific application requirements, the level of real-time performance needed, familiarity with the platforms, the available tools, ecosystem, and the level of determinism needed. **END** 

References

- https://wiki.linuxfoundation.org/realtime/preempt_rt_versions
- <https://mirrors.edge.kernel.org/pub/linux/kernel/projects/rt/>
- <https://bootlin.com/doc/training/preempt-rt/preempt-rt-slides.pdf>
- <https://ubuntu.com/blog/what-is-real-time-linux-part-iii>
- https://elinux.org/images/b/ba/Elc2013_Rostedt.pdf
- https://wiki.archlinux.org/title/Realtime_kernel_patchset
- https://re.public.polimi.it/retrieve/e0c31c12-9844-4599-e053-1705fe0aef77/11311-1076057_Reghenzani.pdf
- <https://static.lwn.net/lwn/images/conf/rtlws11/papers/proc/p19.pdf>
- <https://www.smile.eu/en/white-paperswhite-papers/linux-rt>
- <https://dl.acm.org/doi/abs/10.1145/3297714>
- https://elinux.org/images/d/de/Real_Time_Linux_Scheduling_Performance_Comparison.pdf
- http://www.qnx.com/developers/docs/qnxcar2/index.jsp?topic=%2Fcom.qnx.doc.neutrino.sys_arch%2Ftopic%2Fkernel.Scheduling_latency.html
- http://www.qnx.com/developers/docs/qnxcar2/index.jsp?topic=%2Fcom.qnx.doc.neutrino.sys_arch%2Ftopic%2Fkernel.Interrupt_latency.html
- https://www.mathworks.com/products/connections/product_detail/qnx-neutrino-rtos.html
- <https://blackberry.qnx.com/en/products/foundation-software/qnx-rtos>
- <http://home.iitj.ac.in/~saurabh.heda/Papers/Survey/Report%20on%20%20QNX%20-2007.pdf>

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July 2023	Database management and Optimisation
August 2023	Mobile/Web App Development, Optimisation and Security
September 2023	DevOps Special
October 2023	Blockchain and Open Source
November 2023	Open Source and IoT and Edge
December 2023	All About Data Management
January 2024	Containers and Managing Containers
February 2024	Open Source on Windows and Best in the world of Open Source (Tools and Services)